Explain e-business. Write the advantage and disadvatage of e-commerce.

E-business (Electronic Business) is a broader term than e-commerce. It refers to the use of the internet, intranets, extranets, and other networks to support every part of a business. This doesn't just include buying and selling, but also servicing customers, collaborating with business partners, and conducting electronic transactions within an organization (like HR or finance).

Think of **E-commerce** as the front-end (the shop), while **E-business** is the entire building, including the warehouse, the office, and the communication lines.

E-commerce: Advantages and Disadvantages

E-commerce specifically focuses on the buying and selling of goods and services over the internet.

Advantages of E-commerce

- ✓ **Global Reach:** A business is no longer limited by geography. A small shop in **Nepal** can sell handmade goods to a customer in the **USA**.
- ✓ **24/7 Availability:** Unlike a physical store with "Open" and "Closed" hours, an e-commerce site is always open for business, even while the owner is sleeping.
- ✓ **Lower Operational Costs:** There is no need to pay for a high-street physical store, expensive interior decor, or a large number of on-site staff.
- ✓ **Personalization:** Websites can track user preferences and show them products they are actually interested in, making the shopping experience better.
- ✓ **Convenience:** Customers can shop from the comfort of their homes without having to deal with traffic or crowds.

Disadvantages of E-commerce

- ✓ **Security Issues:** The biggest concern is the risk of credit card fraud, identity theft, or data breaches during online transactions.

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- ✓ **Lack of Physical Touch:** Customers cannot touch, feel, or try on products (like clothes or shoes) before buying, which can lead to high return rates.
- ✓ **Shipping & Delays:** Even if you buy an item in seconds, you have to wait days or weeks for it to arrive. Shipping costs can also make products more expensive.
- ✓ **High Competition:** Since it's easy to start an online shop, the market is crowded. It is very hard for small businesses to compete with giants like Amazon or Alibaba.
- ✓ **Technical Failures:** If the website crashes or the internet goes down, the business stops completely. A physical shop can still operate during a power cut.

What is business model? Explain its types with suitable

A **Business Model** is a company's core strategy for making a profit. It defines what products or services the business will sell, who its target market is, and what expenses it will incur. In simpler terms, it is a "blueprint" that explains how a company creates, delivers, and captures value.

Types of Business Models

In the context of E-commerce and modern business, models are generally categorized based on who is selling to whom.

1. B2B (Business-to-Business)

In this model, one business sells products or services to another business. The consumer is not an individual person but another company.

Example: A software company selling an inventory management system to a local hospital, or a manufacturer selling raw materials to a factory.

2. B2C (Business-to-Consumer)

This is the most common model where a business sells directly to the end-user (the general public).

Example: You buying a laptop from an online store or ordering food through an app.

3. C2C (Consumer-to-Consumer)

In this model, consumers sell directly to other consumers. The platform usually acts as a facilitator and takes a small commission.

Example: Selling your old textbook to another student on **Hamrobazaar** or eBay.

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4. C2B (Consumer-to-Business)

Here, an individual creates value or a product, and a business consumes that value.

Example: An influencer charging a brand to promote a product, or a freelance developer selling a custom script to a tech firm on Fiverr.

5. G2B / B2G (Government-to-Business)

This involves transactions between the government and the private sector.

Example: A construction company winning a government contract to build a new bridge (B2G), or a business paying its yearly taxes through a government portal (G2B).

Other Modern Business Models

Subscription Model: Customers pay a recurring fee (monthly/yearly) to access a service.

Example: Netflix, Spotify, or Microsoft 365.

Freemium Model: Basic services are provided for free, but users must pay for "premium" features.

Example: Canva, LinkedIn, or many mobile games.

Marketplace Model: A platform that brings buyers and sellers together without owning any inventory.

Example: Daraz or Amazon.

What is I-way? Describe the components of I-way What are the types of software agent.

I-way is short for the **Information Superhighway**. It is a high-speed communication network that delivers text, graphics, video, and other digital data to homes and businesses. Essentially, it is the underlying infrastructure (the "pipes") that makes e-commerce and digital communication possible.

Think of it like a national highway system, but instead of cars and trucks, it carries **data packets**.

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2. Components of I-way

The Information Superhighway is not a single cable; it is built of three major components that allow data to travel from a provider to your screen:

Consumer Access Equipment: These are the devices used by the end-user to access the network.

Examples: Personal Computers (PCs), Smartphones, Set-top boxes, and Modems.

Local On-Ramps (Access Links): These are the "entry points" or the connection between your home/office and the main backbone of the internet.

Examples: Fiber-optic cables (FTTH), DSL (Digital Subscriber Line), Cable TV lines, and Wireless links (5G/4G).

Global Information Distribution Network (The Backbone): This is the long-distance, high-capacity network that connects different cities and countries. It is the "main highway" of the internet.

Examples: Undersea fiber-optic cables and Satellites.

3. Types of Software Agents

A **Software Agent** (or "Bot") is a program that acts on behalf of a user to perform repetitive or complex tasks autonomously. In the world of E-commerce and Data Mining, there are several types:

A. Buyer Agents (Shopping Bots)

These agents help users find products and services. They can compare prices from different websites and alert the user when a deal is found.

Example: An agent that monitors airline ticket prices and notifies you when they drop.

B. User Agents (Personal Agents)

These agents perform tasks for a specific user to save time.

Example: An agent that automatically sorts your emails into folders or fills out web forms with your information.

C. Monitoring and Surveillance Agents

These agents observe and report on computer systems and networks.

Example: An agent that monitors a website's "uptime" and alerts the admin if the site goes down.

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D. Data Mining Agents

These agents use AI and pattern recognition to find trends in large databases.

Example: An agent used by a bank to detect suspicious credit card transactions that look like fraud.

E. Distributed Agents

These agents work in a group to solve a problem. They communicate with each other to reach a goal.

Example: Agents used in complex supply chain management to coordinate between suppliers and manufacturers.

Define EDI and its applications. Explain security and privacy issues EDI.

1. EDI

EDI (Electronic Data Interchange) is the computer-to-computer exchange of business documents in a standard electronic format between business partners.

Traditionally, businesses exchanged documents like purchase orders and invoices using paper or email (which still requires manual entry). EDI replaces these with a structured digital format that allows systems to "talk" to each other without human intervention.

2. Applications of EDI

EDI is used across various industries to speed up transactions and reduce errors:

- ✓ **Inventory Management:** Retailers like Walmart use EDI to automatically send "Restock" orders to suppliers when shelves are low.
- ✓ **Logistics & Shipping:** To track shipments, send Bills of Lading, and provide delivery status updates in real-time.
- ✓ **Financial Services:** Used for Electronic Funds Transfer (EFT) and automated payment processing between banks and corporations.
- ✓ **Healthcare:** Exchanging patient records, insurance claims, and prescriptions securely between hospitals and insurance providers.

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- ✓ **Manufacturing:** Coordinating "Just-in-Time" (JIT) manufacturing by syncing parts delivery from suppliers with the production schedule.

3. Security and Privacy Issues in EDI

Since EDI involves sensitive financial and corporate data, it faces several security challenges:

A. Data Confidentiality (Privacy)

The main concern is that unauthorized parties might intercept business secrets or financial data during transmission.

Solution: Use of strong **Encryption** (like SSL/TLS) to make the data unreadable to hackers.

B. Data Integrity

If an EDI message (like a purchase order for 10 units) is altered by a hacker to say 1,000 units, it could cause huge financial loss.

Solution: Using **Digital Signatures** and checksums to ensure the message hasn't been changed during transit.

C. Authentication & Non-Repudiation

Businesses need to be 100% sure that a message actually came from their partner and not an impostor. **Non-repudiation** ensures that a sender cannot later deny sending a message.

Solution: Using **Digital Certificates** and secure Value-Added Networks (VANs).

D. System Vulnerability

Since EDI systems are often directly linked to a company's internal database (like ERP systems), a breach in the EDI entry point could give a hacker access to the entire company's server.

Solution: Implementing **Firewalls** and strict access control layers.

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Define m-Commerce. Write mobile commerce architectures.

M-Commerce (Mobile Commerce) refers to any monetary transaction that is completed using a mobile telecommunications network and a handheld device, such as a smartphone or a tablet.

Essentially, it is the delivery of electronic commerce capabilities directly into the consumer's hand, anywhere, via wireless technology.

Mobile Commerce Architecture

The architecture of m-commerce is a multi-layered structure that ensures a request from a mobile phone reaches the business server and returns the correct information securely.

1. The 3-Tier Architecture

Most m-commerce systems follow a three-tier model:

- ✧ Tier 1: Presentation Layer (Client): The user's mobile device (App or Browser). This is where the user interacts with the interface.
- ✧ Tier 2: Application Layer (Middleware): This is the server that processes the logic. It handles security, transaction management, and communication between the mobile device and the database.
- ✧ Tier 3: Data Layer (Database): The backend where product information, user profiles, and transaction records are stored.

2. Network Infrastructure Components

To make the architecture work, several network elements are involved:

- ✓ Mobile Station (MS): Your smartphone or tablet.
- ✓ Base Transceiver Station (BTS): The mobile tower that receives the wireless signal from your phone.
- ✓ Mobile Switching Center (MSC): The "brain" of the mobile network that routes calls and data to the correct destination.
- ✓ Gateway: A bridge that connects the mobile network (like 5G/LTE) to the Public Internet.

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- ✓ WAP Gateway (Wireless Application Protocol): A specialized component that translates internet data into a format that mobile devices can easily display.

Key Features of M-Commerce Architecture

- ✓ Ubiquity: Users can perform transactions from anywhere (home, bus, office).
- ✓ Reachability: Businesses can reach customers instantly via push notifications.
- ✓ Localization: Using GPS, the architecture can provide services based on the user's current location (e.g., finding the nearest ATM).
- ✓ Personalization: Mobile devices are personal, allowing the architecture to store unique user preferences.

What is cryptography and explain its types. Differentiate between digital certificate and digital signature.

Cryptography is the science of securing information by transforming it into an unreadable format. Its primary goal is to ensure that only the intended recipient can read and process the data, protecting it from unauthorized access, alteration, or theft.

In the world of E-commerce and Computer Engineering, cryptography provides the "lock and key" for sensitive data like passwords, credit card numbers, and private messages.

Types of Cryptography

There are two main types of cryptography based on the "keys" used to encrypt and decrypt the data:

1. Symmetric Key Cryptography (Secret Key)

In this type, the **same key** is used for both encryption and decryption. Both the sender and the receiver must have a copy of this private key.

- ➢ Pros: Very fast; ideal for encrypting large amounts of data.
- ➢ Cons: Key distribution is a problem (how do you send the key to the other person safely?).
- ➢ Example: AES (Advanced Encryption Standard), DES.

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2. Asymmetric Key Cryptography (Public Key)

This type uses a pair of keys: a Public Key (which can be shared with anyone) and a Private Key (which is kept secret by the owner).

If data is encrypted with the Public Key, it can *only* be decrypted by the corresponding Private Key.

Pros: Solves the key distribution problem; very secure.

Cons: Slower than symmetric encryption due to complex math.

Example: RSA, Diffie-Hellman.

Difference between Digital Certificate and Digital Signature

Feature	**Digital Signature**	**Digital Certificate**
Purpose	To verify the authenticity and integrity of a document (proves it wasn't changed).	To verify the identity of the sender (proves who you are).
What it does	It "signs" a specific message or file.	It "binds" a public key to an individual or organization.
Analogy	Like a handwritten signature on a contract.	Like an ID Card or a Driver's License.
Issued By	The sender themselves (using their private key).	A trusted third party called a Certificate Authority (CA).
Mechanism	Uses hashing and the sender's private key.	Contains the owner's public key and info about the CA.

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What are the e-commerce security issues. Explain the types and function of firewall.

In E-commerce, security is the top priority because it involves the exchange of money and sensitive personal information. Any breach can lead to financial loss and a total loss of customer trust.

1. E-commerce Security Issues

Security issues in e-commerce can be categorized into several common threats:

- ✓ **Phishing:** Attackers create fake websites that look like legitimate ones (e.g., a fake bank login page) to steal user credentials and credit card details.
- ✓ **Data Breaches:** Unauthorized access to a company's database to steal customer information, such as names, addresses, and payment history.
- ✓ **Denial of Service (DoS/DDoS):** Flooding a website with so much traffic that it crashes, preventing real customers from making purchases.
- ✓ **Man-in-the-Middle (MITM) Attacks:** An attacker intercepts the communication between the customer and the server to steal or alter data during the transaction.
- ✓ **Malware & Ransomware:** Malicious software that can infect a user's device or a company's server to steal data or lock it until a ransom is paid.

Functions of a Firewall:

- ✓ **Packet Filtering:** It inspects every "packet" of data entering or leaving the network and blocks those that do not meet security criteria.
- ✓ **Access Control:** It prevents unauthorized users from accessing private networks.
- ✓ **Content Filtering:** It can block access to specific websites or types of content (e.g., blocking social media in an office).
- ✓ **Monitoring & Logging:** It keeps a record of all traffic, allowing administrators to see if there have been any attempted attacks.

3. Types of Firewalls

There are several ways to implement firewalls depending on the level of security required:

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I. Packet Filtering Firewall

The most basic type. It checks each packet's source and destination IP addresses, protocol, and port numbers. If the packet doesn't match the "allow" list, it is dropped.

- **Pros:** Fast and efficient.
- **Cons:** Cannot inspect the actual data *inside* the packet.

II. Application-Level Gateway (Proxy Firewall)

This firewall operates at the **Application Layer**. It acts as an intermediary (proxy) between the user and the server.

- **Function:** It inspects the actual content of the data (like HTTP or FTP requests).
- **Pros:** Very secure because it masks your internal network identity.

III. Stateful Inspection Firewall

This is more advanced than packet filtering. It doesn't just look at individual packets; it tracks the **state** of active connections.

Function: It remembers if a connection was already established and only allows packets that are part of that valid conversation.

IV. Next-Generation Firewall (NGFW)

A modern firewall that combines traditional filtering with advanced features like **Deep Packet Inspection (DPI)**, encrypted traffic inspection, and an Integrated Intrusion Prevention System (IPS).

Differentiate between entrepreneur and manager. Explain the characteristics of successful entrepreneur.

1. Difference between Entrepreneur and Manager

Feature	Entrepreneur	Manager
Primary Goal	To innovate and create a new venture.	To manage and sustain an existing business.
Status	The owner of the enterprise.	An employee of the enterprise.
Risk Bearing	Bears all the financial and personal risks.	Does not bear any financial risk.

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Feature	Entrepreneur	Manager
Reward	Profit (can be unlimited).	Fixed Salary and occasional bonuses.
Innovation	Constantly looks for new ideas/changes.	Follows established policies and rules.
Motivation	Achievement and independence.	Power and job security.

2. Characteristics of a Successful Entrepreneur

- ✓ **Risk-Taking Ability:** They are willing to step into the unknown. They don't take "blind" risks but rather **calculated risks** where they have analyzed the potential for success.
- ✓ **Innovation & Creativity:** Entrepreneurs don't just sell products; they solve problems. They look for new ways to improve existing services or create entirely new markets.
- ✓ **Self-Motivation & Discipline:** Since they are their own boss, they must have the drive to work long hours without anyone telling them to do so.
- ✓ **Vision & Goal Orientation:** They have a clear picture of where they want their business to be in 5 or 10 years. They focus on the "big picture."
- ✓ **Resilience (Persistence):** Failure is common in new ventures. A successful entrepreneur views failure as a learning opportunity and keeps moving forward.
- ✓ **Decision-Making Skills:** They must be able to make quick, firm decisions, often with limited information.
- ✓ **Flexibility:** Markets change fast. Successful entrepreneurs are "agile"—they can pivot their business model when they see that their current path isn't working.

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Write short notes:

A. Digital wallet

A **Digital Wallet** (also known as an **e-Wallet**) is a software-based system that securely stores a user's payment information and passwords for numerous payment methods and websites. It allows users to make electronic transactions quickly and easily using a smartphone or a computer.

In the context of Nepal, examples include platforms like **eSewa**, **Khalti**, and **IME Pay**, which have revolutionized how people pay for utilities, groceries, and travel.

How it Works

A digital wallet uses various technologies to facilitate transactions:

- ✓ **Storage:** It stores virtual versions of debit/credit cards or maintains a "stored value" (pre-paid balance).
- ✓ **Communication:** It uses technologies like **NFC (Near Field Communication)** for contactless payments, **QR Codes** for scanning, or secure internet protocols for online shopping.
- ✓ **Security:** Transactions are usually protected by encryption, biometrics (fingerprint/Face ID), and **PINs/OTPs**.

Key Components

Software: The actual application installed on the device.

Information: Encrypted data including credit/debit card numbers, shipping addresses, and loyalty card info.

Payment Infrastructure: The linked banking network that processes the actual movement of money.

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B. SSL

SSL (Secure Sockets Layer)

SSL is the standard security technology for establishing an encrypted link between a web server and a browser. This link ensures that all data passed between the web server and browsers remain private and integral.

In modern systems, SSL has been succeeded by TLS (Transport Layer Security), though most people still refer to it as "SSL." It is the technology that puts the "S" in HTTPS and displays the Padlock icon in your browser's address bar.

How SSL Works (The SSL Handshake)

The process of establishing a secure connection is called the **SSL Handshake**, which uses **Asymmetric Cryptography** to create a secure session.

- ✓ Browser Connects: The browser attempts to connect to a website secured with SSL.
- ✓ Server Sends Certificate: The server sends its SSL Certificate (which includes its Public Key) to the browser.
- ✓ Browser Checks: The browser checks if the certificate is from a trusted Certificate Authority (CA) and hasn't expired.
- ✓ Session Key Creation: The browser creates a unique, symmetric Session Key, encrypts it with the server's Public Key, and sends it back.
- ✓ Encryption Begins: The server decrypts the Session Key using its Private Key. Now, both parties use this same key to encrypt all transmitted data (Symmetric Encryption).

importance of SSL in E-commerce

- ✓ Data Encryption: Protects sensitive info like passwords and credit card numbers from "sniffing" or interception.
- ✓ Authentication: Ensures the user is talking to the *real* website and not a "spoofed" fake site.
- ✓ Data Integrity: Ensures that data is not altered during transit (e.g., changing the amount in a bank transfer).
- ✓ SEO & Trust: Google gives higher rankings to HTTPS websites, and customers are more likely to buy if they see the padlock icon.

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C. Aims of cyber law

In the context of your **Computer Engineering** and **E-commerce** studies, **Cyber Law** (also known as Internet Law) is the legal framework that governs the use of the internet and digital technology.

The primary "Aims" (Goals) of Cyber Law are to protect users and businesses while promoting a secure digital economy.

Primary Aims of Cyber Law

1. Legal Recognition of Digital Records

Before cyber laws, only paper documents with physical signatures were considered legal. Cyber law aims to give **Electronic Data Interchange (EDI)**, digital signatures, and electronic contracts the same legal status as their paper counterparts.

2. Protection of Intellectual Property (IP)

Cyber law aims to protect the creations of the mind in the digital space. This includes:

- ✓ **Copyright:** Protecting software code, digital music, and articles.
- ✓ **Trademarks:** Preventing others from using a company's brand name or logo online.
- ✓ **Patents:** Protecting unique technical inventions in hardware or software.

3. Prevention of Cyber Crimes

One of the most critical aims is to define what constitutes a crime in the digital world and set punishments for them. This includes:

- ✓ **Hacking:** Unauthorized access to computer systems.
- ✓ **Data Theft:** Stealing sensitive corporate or personal data.
- ✓ **Cyber Terrorism:** Attacks on a nation's critical infrastructure (power grids, banking).
- ✓ **Phishing & Fraud:** Online financial scams.

4. Ensuring Data Privacy and Security

Cyber law mandates how companies must handle and protect user data. It aims to prevent the misuse of personal information and ensures that businesses implement strong security measures like **Firewalls** and **Encryption**.

5. Regulation of E-commerce

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It provides a safe environment for online business. It ensures that:

- ✓ Online contracts are binding.
- ✓ Consumers are protected from fraudulent sellers.
- ✓ Electronic payment systems (like digital wallets) are regulated.

D. Online marketing VS Offline marketing

Online Marketing

Online marketing (also called digital marketing) refers to promoting products or services using the internet and digital technologies.

Examples:

- Social media marketing (Facebook, Instagram)
- Email marketing
- Search engine optimization (SEO)
- Online ads (Google Ads)

Advantages:

- Global reach (can target worldwide audience)
- Cost-effective compared to traditional methods
- Easy to track and measure results (analytics)
- Fast communication and instant feedback

Disadvantages:

- Requires internet access
- High competition
- Security and privacy issues
- Can be ignored easily (ad blockers, spam)

Offline Marketing

Offline marketing refers to traditional methods of promotion that do not involve the internet.

Examples:

- TV and radio advertisements
- Newspapers and magazines

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- Billboards and posters
- Flyers and brochures

Advantages:

- More trustworthy for some audiences
- Tangible (physical materials like flyers)
- Effective for local targeting
- Less dependence on technology

Disadvantages:

- Expensive (printing, broadcasting costs)
- Limited reach (mostly local or regional)
- Difficult to measure performance
- Slow communication and feedback

Feature	**Online Marketing**	**Offline Marketing**
Medium	Websites, Social Media, Email, SEO, Apps.	TV, Radio, Newspapers, Billboards, Flyers.
Reach	Global: Can reach anyone in the world instantly.	Local/Regional: Limited by physical distribution.
Cost	Generally Cheaper (can start with a small budget).	Expensive (TV ads and Billboards cost a lot).
Targeting	High: Can target specific ages, interests, and locations.	Low: Reaches a mass audience with little control over who sees it.
Tracking	Real-time: You can see exactly how many people clicked or bought.	Delayed/Difficult: Hard to know exactly how many people bought because of a specific flyer.
Interaction	Two-way: Customers can comment, like, and chat instantly.	One-way: The audience just listens or looks; they cannot respond directly.

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